CS-1101
PROGRAMMING FUNDAMENTALS

Lab 14

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Lab Contents

- Structures
- Arrays of Structures
- Structure Pointers

Lab Objectives

The objective of this lab is to apply the:

- concept of structures in C++.
- concept of arrays of structures.
- concept of structure pointers.

Structures

- A structure is a collection of **variables of different data types** and **member functions** under a single name.
- Suppose you want to store some information about a person: their **first_name, last_name, age, and salary**. You can easily create different variables—first_name, last_name, age, salary to store this information separately.
- For multiple people, you'd need **first_name1, last_name1, age1, salary1, first_name2, last_name2, age2, salary2, etc.** The number of variables grows quickly, making the code messy.
- A better approach is to have a collection of all related information under a single name, such as **Person** and use it for every individual.

How to Declare a Structure in C++

- A structure is a user-defined data type in C/C++. A structure creates a data type that can be used to group **items of possibly different types** into a single type.

Keyword

Identifier

```
struct Person{
    int age;
    string first_name;
    string last_name;
    int salary;
};
```

Data Members

How to Define a Structure Variable in C++

- There two ways to define a structure variable.

```
struct Person{
    int age;
    string first_name;
    string last_name;
    int salary;
} p1, p2;
```

Both p1 and p2 are variable of type Person.

```
int main(){

    Person p1;

    return 0;
}
```

Here p1 is a variable of type Person.

Member Functions in Structures

- Structures can also have member functions.
- These member functions are similar to regular functions but are defined within the scope of a structure. They can access and manipulate the data members of the structure directly.
- We can declare a member function by defining the function within the structure definition.

```
int main(){
//  First Person
    Person p1;
    p1.age = 30;
    p1.first_name = "Muhammad";
    p1.last_name = "Awais";
    p1.salary = 30000;
    p1.display(p1);
//  Second Person
    Person p2;
    p2.first_name = "Kashif";
    p2.last_name = "Hameed";
    p2.age = 34;
    p2.salary = 50000;
    p2.display(p2);
    return 0;
}
```

Calling a member function.

```
#include<iostream>
using namespace std;
struct Person{
    int age;
    string first_name;
    string last_name;
    int salary;
    void display(Person person){
        cout << "Name : " <<  person.first_name << " " << person.last_name << endl;
        cout << "Age : " << person.age << endl;
        cout << "Salary : " << person.salary << endl;
    }
}p1,p2;
```

Member Functions in C++

Array of Structures

- Like other primitive data types, we can create an array of structures.

```
int main(){
//  First Person
    Person p1;
    p1.age = 30;
    p1.first_name = "Muhammad";
    p1.last_name = "Awais";
    p1.salary = 30000;
    p1.display(p1);
//  Second Person
    Person p2;
    p2.first_name = "Kashif";
    p2.last_name = "Hameed";
    p2.age = 34;
    p2.salary = 50000;
    p2.display(p2);

//  Array of Structures
    Person persons[2] = {p1,p2};
    for (int i =0 ; i < 2 ; i++){
        persons[i].print();
    }
    return 0;
}
```

Structure Pointers

- Like primitive types, we can have pointer to a structure. If we have a pointer to structure, members are accessed using arrow (->) operator instead of the dot (.) operator.

```
int main(){
    Person p2;
    p2.first_name = "Kashif";
    p2.last_name = "Hameed";
    p2.age = 34;
    p2.salary = 50000;
    p2.display(p2);
//  Pointer to a structure
    Person *personPtr = &p2;
    cout << "Print name with pointer : " << personPtr->first_name << endl;
    return 0;
}
```

Tasks

- Declares a structure named Student to store:
  - First Name
  - Last Name
  - Roll Number
  - Marks
- Includes a member function displayStudentInfo() to:
  - Display the full name of the student.
  - Display the marks of the student.
- Uses a structure variable to:
  - Assign values to the data members.
  - Call the member function to display the student's information.

Tasks

- Repeat the previous program, but this time:
  - Use an array of structures to store details of multiple students.
  - Allow the user to input details for multiple students.
  - Call the displayStudentInfo() function for each student to display their full name and marks.
- Modify your previous program to:
  - Use a pointer to a structure to store and access student details.
  - Dynamically assign values to the structure members using the pointer.
  - Use the pointer to call the displayStudentInfo() function

Home Tasks

- What is the purpose of using **this** in member functions of structure?

References

- https://www.geeksforgeeks.org/structures-in-cpp/